

Validation Summary: 100 One-Day Lunar Orbit Propagation Cases

Model	Median RMS [m]	P95 RMS [m]	Max RMS [m]	Total Runtime [s]	Per-Scenario Runtime [s]	Speedup vs Truth
GPU SH20 RK4	98.510	389.685	571.026	1714.6	17.15	2.5x
GPU ST-LRPS RK4	0.158	0.689	0.960	1894.6	18.95	2.2x
GPU SH60 RK4	0.058	1.199	2.299	4897.4	48.97	0.9x